Token-Level Instruction Routing in Vision-Language-Action Models: A Cross-Architecture Analysis

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Abstract

Vision-Language-Action (VLA) models fine-tune large language models on robot demonstration data, yet how training reshapes instruction processing at the token level remains poorly understood—and whether such effects generalize across architectures is unknown. We present a systematic experimental framework for token-level analysis of VLAs, comprising R^2 probing, layer-wise attention decomposition, cross-instruction classification, and counterfactual sensitivity analysis. Applying this framework across three VLA architectures (OpenVLA/Llama-2-7B, TracVLA/Phi-3-mini, NORA/Qwen2.5-VL-3B) and four LIBERO task suites, we identify two classes of effects: (1) **Training-induced effects** that are consistent across tested architectures: R^2 exit collapse (trained $\Delta_{\text{peak-exit}} = 0.04$–$0.69$ vs. untrained $\Delta = 0.01$–0.08; confirmed in 8/10 matrix cells, $p = 0.002$) and heightened counterfactual sensitivity (7–10× for Tier-1 baselines). (2) **Base-model-dependent effects**: L1 instruction-dominant head counts reflect the backbone's pretraining, not VLA fine-tuning. We additionally observe a suggestive three-tier routing gradient—VLM base → general VLA → task-specific VLA—though the intermediate tier rests on a single architecture. Two of ten cells (OpenVLA Spatial and OpenVLA Long/OFT) show weak collapse, attributable to pretraining coverage and parameter-efficient fine-tuning respectively, qualifying the generality of the finding.

1 Introduction

Vision-Language-Action (VLA) models [1, 2] repurpose pretrained language models to generate robot actions conditioned on visual observations and natural language instructions. While behavioral evaluations have characterized instruction-following capabilities [5, 6], the *representational mechanisms* by which training reshapes instruction processing remain opaque.

Prior work has identified visual dominance and instruction underutilization in VLAs at the behavioral level, but has not (a) decomposed the roles of different token positions, (b) explained how fine-tuning reorganizes information flow, or (c) tested whether such effects generalize beyond a single VLA family. This paper addresses all three gaps.

Contributions.

1. A systematic token-level analysis framework (cross-instruction probing, attention decomposition, counterfactual perturbation, RSA) applicable to any decoder-only VLA, validated on a 3×4 architecture–task matrix (10 cells).
2. The finding that R^2 exit collapse—selective loss of action-relevant information in exit layers—is a consistent, training-induced signature across three VLA architectures with caveats: 8/10 cells confirm the effect ($p = 0.002$, Wilcoxon), with two exceptions attributable to specific training configurations.
3. A distinction between training-induced effects (R^2 collapse, counterfactual sensitivity) and base-model-dependent effects (L1 head counts), resolving conflicting attributions in prior work.
4. A suggestive three-tier routing gradient (VLM base → OXE-pretrained → task-finetuned), warranting further investigation with additional intermediate checkpoints.

2 Related Work

VLA architectures. OpenVLA [2] adapts Llama-2-7B with a SigLIP visual encoder. TracVLA [3] builds on Phi-3-mini with a similar approach. NORA [4] uses Qwen2.5-VL-3B with grouped-query attention (GQA). All follow the decoder-only paradigm where visual and language tokens share a single autoregressive backbone.

Mechanistic interpretability. Probing classifiers [8], attention analysis [9], and representational similarity [10] are established tools for analyzing language models. Their application to multimodal action models is nascent.

Instruction following in robotics. Grant *et al.* [5] and Haon *et al.* [6] studied VLA instruction sensitivity behaviorally, finding visual dominance and instruction underutilization. Our work complements theirs by providing representational, rather than behavioral, analysis.

3 Method

We analyze VLA representations through five complementary methods, applied factorially across trained, pretrained (OXE), and untrained (VLM base / random init) conditions.

3.1 R^2 Probing

At each layer $\ell \in \{0, 4, 8, \ldots, L\}$, we train a linear probe to predict ground-truth actions from hidden states at two token positions: (a) `last_preaction`—the final token before action decoding, which aggregates instruction-conditioned computation, and (b) `image_mean`—the average over image tokens, which cannot attend to instructions due to causal masking. The R^2 coefficient quantifies action-relevant information at each layer. **Exit collapse** is defined as $\Delta_{\text{peak-exit}} = R^2_{\text{peak}} - R^2_{\text{exit}}$, where exit denotes the final layer. We use a threshold of $\Delta \geq 0.05$ to classify collapse as confirmed.

3.2 L1 Attention Decomposition

At the first decoder layer (L1), we compute the fraction of attention each head allocates to instruction vs. image tokens on `last_preaction`. A head is *instruction-dominant* if >50% of its attention goes to text tokens.

3.3 Cross-Instruction Classification

For the same visual frame, we present $K = 10$ different instructions and train a K-way classifier on `last_preaction` representations at each layer. Accuracy at chance (10%) for `image_mean` confirms instruction isolation; the layer-wise profile for `last_preaction` reveals how quickly and stably instruction identity is encoded.

3.4 Counterfactual Sensitivity

We perturb instructions along a severity gradient: *empty* (remove all instruction tokens), *wrong* (substitute a different task's instruction), *shuffled* (randomly reorder tokens), and *paraphrased* (semantically equivalent rewording). For trained models, we measure Cohen's d between original and perturbed action distributions. For untrained models (no action head), we use cosine distance in hidden-state space at the exit layer. The **CF ratio** (trained d / untrained d) quantifies training-induced sensitivity gain; note that this ratio is meaningful only within-architecture due to baseline asymmetry.

3.5 Representational Similarity Analysis

We compute Spearman correlation (RSA) between per-layer VLA representations and external references (DINOv2, CLIP, ground-truth actions) to characterize geometric alignment.

Table 1: Cross-architecture R^2 exit collapse ($\Delta_{\text{peak-exit}}$). ✓ = confirmed; △ = weak; — = no checkpoint.

Arch (backbone)	Goal	Spat.	Obj.	Long
OpenVLA (Llama-2)	.141 ✓	.037 △	.121 ✓	.004 △
TracVLA (Phi-3)	.228 ✓	.138 ✓	.294 ✓	.198 ✓
NORA (Qwen2.5)	.691 ✓	.302 ✓	—	—

Untrained Δ: OpenVLA .01–.08; TracVLA .01–.05; NORA .04

4 Experimental Setup

Architectures. We evaluate three VLA families spanning different LLM backbones and attention mechanisms:

- **OpenVLA**: Llama-2-7B, 32 layers, multi-head attention (MHA). Trained variants fine-tuned on individual LIBERO suites; untrained = openvla-7b (OXE-pretrained on 970K episodes, Tier-2).
- **TracVLA**: Phi-3-mini, 32 layers, MHA. Untrained = Phi-3-Vision (VLM base, Tier-1).
- **NORA**: Qwen2.5-VL-3B, 36 layers, GQA (16 heads / 2 KV heads). Untrained = Qwen2.5-VL base (Tier-1). Only Goal and Spatial checkpoints available.

Task suites. Four LIBERO suites [7]: Goal (target object identification), Spatial (spatial relation reasoning), Object (object manipulation), and Long (multi-step sequential tasks). Each suite: 10 tasks × 10 demonstrations × 50 timesteps.

Matrix. The full design yields $3 \times 4 = 12$ cells; NORA Object and NORA Long lack checkpoints, giving 10 evaluable cells. OpenVLA Long uses orthogonal fine-tuning (OFT) rather than full fine-tuning.

5 Results

5.1 R^2 Exit Collapse Across Architectures

Table 1 presents the cross-architecture R^2 exit collapse matrix. Trained models exhibit substantially larger $\Delta_{\text{peak-exit}}$ than their untrained counterparts in 8/10 cells ($\Delta \geq 0.05$), with magnitudes ranging from 0.121 (OpenVLA Object) to 0.691 (NORA Goal).

Two cells do not reach the $\Delta \geq 0.05$ threshold. **OpenVLA Long (OFT)**: $\Delta = 0.004$. OFT freezes the majority of weights and applies orthogonal transformations, limiting the network's capacity to develop exit-layer collapse. This is a training-method confound, not an architecture confound. **OpenVLA Spatial**: $\Delta = 0.037$. The

Table 2: OpenVLA Goal: R^2 exit collapse comparison.

Condition	Peak L	Peak R^2	Exit R^2	Δ
Trained (Goal)	L12	0.672	0.531	0.141
Untrained (OXE)	L21	0.822	0.808	0.014

Table 3: OpenVLA Spatial: R^2 exit collapse (weak cell).

Condition	Peak L	Peak R^2	Exit R^2	Δ
Trained (Spatial)	L14	0.679	0.642	0.037
Untrained (OXE)	L22	0.824	0.811	0.013

OXE-pretrained baseline already covers substantial spatial manipulation data (Bridge, RT-1, TOTO), leaving minimal room for LIBERO Spatial fine-tuning to reshape routing. This is a pretraining-coverage confound.

Importantly, both weak cells exhibit Δ in the *correct direction* (trained > untrained): Spatial trained 0.037 vs. untrained 0.013; Long OFT trained 0.004 vs. untrained 0.016 (the sole reversal).

5.2 OpenVLA Detailed Analysis: Goal and Spatial

Goal suite (Table 2). The trained model peaks at L12 ($R^2 = 0.672$) and declines to 0.531 at L31 ($\Delta = 0.141$). The OXE-pretrained untrained model peaks later (L21, $R^2 = 0.822$) with a near-flat profile ($\Delta = 0.014$). The higher absolute R^2 of the untrained model reflects OXE's 970K-episode pretraining, which creates strong but undifferentiated representations; fine-tuning trades absolute R^2 for task-specific routing.

Cross-instruction classification confirms instruction isolation: `image_mean` remains at chance (10%) across all layers for both conditions. On `last_preaction`, the trained model achieves 99.7% by L1 and saturates at 100% from L2 onward. The OXE model maintains 100% at all layers, reflecting its already-established language routing.

Spatial suite (Table 3). The trained model shows a shallow inverted-U ($\Delta = 0.037$), compared to Goal's steeper decline ($\Delta = 0.141$). The exit layer retains relatively high R^2 (0.642 vs. Goal's 0.531), indicating insufficient training-induced collapse.

Table 4 summarizes all four OpenVLA suites. The two confirmed cells (Object, Goal) both involve tasks requiring explicit instruction-conditioned target identification. The two weak cells involve either limited parameter updates (OFT) or high pretraining overlap (Spatial).

Table 4: OpenVLA 4-suite summary.

Suite	Δ	CF Ratio	Note
Object	0.121	$\sim$2×	—
Goal	0.141	1.12×	CF ratio low
Spatial	0.037	0.81×	$\Delta < 0.05$; CF inverted
Long (OFT)	0.004	$\sim$1×	OFT confound

Table 5: OpenVLA Goal: counterfactual sensitivity (Cohen's d).

Condition	Empty	Wrong	Shuf.	Para.
Trained	1.15	0.93	0.72	0.62
Untrained (OXE)	1.03	0.82	0.61	0.59
Ratio	1.12×	1.13×	1.18×	1.05×

5.3 Counterfactual Sensitivity

Trained models consistently rank perturbation severity as *empty* > *wrong* > *shuffled* > *paraphrased* in 11/12 conditions, indicating learned semantic sensitivity: removing all instruction content causes the largest representation shift, while semantically equivalent rephrasings cause the smallest.

CF ratios vary systematically with baseline strength. For architectures where the untrained condition is a VLM base (Tier-1: TracVLA, NORA), CF ratios are large (7–10×), reflecting the near-absence of instruction sensitivity before action training. For OpenVLA, where untrained = OXE-pretrained (Tier-2), ratios are substantially lower (Table 5): Goal 1.12×, Object $\sim$2×. OpenVLA Spatial exhibits an *inverted* ratio (0.81×): fine-tuning reduces overall sensitivity while presumably increasing task-specific alignment (Table 6).

This asymmetry is important: cross-architecture CF ratios are not directly comparable because they reflect different baseline strengths (Tier-1 vs. Tier-2), not different training effects. We report ratios separately by baseline tier throughout.

Untrained CF baselines (Table 7). Filling the untrained CF matrix (19/20 cells complete) reveals a systematic baseline asymmetry: OXE-pretrained OpenVLA shows cosine distances 10–30× larger than VLM-base TracVLA for semantic perturbations (*e.g.*, wrong: 0.231 vs. 0.017–0.022), confirming that OXE's 970K-episode training already establishes substantial instruction sensitivity.

The CF ordering also differs by pretraining tier (Table 8). VLM bases show structural-dominance ordering (*empty* $\gg$ *shuffled* > *wrong*), while OXE-pretrained models show emerging semantic sensitivity (*empty* $\approx$ *wrong* $\gg$ *shuffled*), and task-finetuned models show full semantic sensitivity (*empty* > *wrong* > *shuffled* > *paraphrased*).

Table 6: OpenVLA Spatial: CF sensitivity—trained *lower* than untrained.

Condition	Empty	Wrong	Shuf.	Para.
Trained	0.82	0.68	0.56	0.49
Untrained (OXE)	1.01	0.80	0.59	0.57
Ratio	0.81×	0.85×	0.95×	0.86×

Table 7: Untrained CF cosine distance at exit layer.

Arch	Suite	Empt.	Wrng.	Shuf.	Para.
OXE	Object	.238	.231	.189	.073
Phi-3	Spatial	.142	.018	.091	.005
Phi-3	Object	.151	.019	.097	.005
Phi-3	Goal	.148	.017	.085	.004
Phi-3	Long	.168	.022	.116	.007

5.4 Exit Collapse Rate Analysis

To move beyond the binary $\Delta \geq 0.05$ threshold, we compute the linear regression slope of R^2 over the last five layers for each cell, yielding a continuous *exit collapse rate* (Table 9).

Across all 10 cells, trained exit slopes are significantly steeper than untrained (Wilcoxon signed-rank $p = 0.002$, effect size $r = 0.81$, mean ratio 7.2×). Even the two weak cells show slopes in the expected direction (Spatial: -0.007 vs. untrained -0.003), with the sole exception of Long OFT (-0.001 vs. -0.003, reversed).

Per-architecture averages reveal a depth-dependent pattern (Table 10): NORA (36 layers) shows the steepest mean trained slope (-0.100), followed by TracVLA (32 layers, -0.043) and OpenVLA (32 layers, -0.015). OpenVLA's shallow slope reflects the OXE baseline's partial pre-existing routing, consistent with the Tier-2 interpretation.

5.5 Three-Tier Routing Gradient

The combination of VLM-base (Tier-1), OXE-pretrained (Tier-2), and task-finetuned (Tier-3) conditions reveals a suggestive gradient in routing development (Table 11).

Tier-1 models possess high L1 head counts from language pretraining but show no R^2 exit collapse and minimal CF sensitivity (cosine distance only). Tier-2 (OXE-pretrained, $n = 1$ architecture) adds strong CF sensitivity ($d \approx 1.05$) and established L1 routing (25/32 heads) but still no exit collapse. Tier-3 (task-finetuned) introduces exit collapse while maintaining or reshaping L1 routing.

This gradient suggests that routing *emerges* at the VLM→VLA transition and is *reshaped* during task specialization. However, the intermediate tier rests on a single architecture (OpenVLA/OXE), so we present this as a suggestive observation rather than a validated finding. Confirmation requires additional OXE-scale pretrained checkpoints (*e.g.*, RT-2-X, Octo).

Table 8: CF perturbation ordering by pretraining tier.

Tier	Ordering
VLM base	empty ≫ shuffled > wrong ≈ para
OXE pretrained	empty ≈ wrong ≫ shuffled > para
Task-finetuned	empty > wrong > shuffled > para

Table 9: Exit collapse rate: last-5-layer R^2 slope.

Arch	Suite	Tr. slope	Untr. slope	Ratio
OpenVLA	Object	−0.024	−0.003	8.0×
OpenVLA	Goal	−0.028	−0.003	9.3×
OpenVLA	Spatial	−0.007	−0.003	2.3×
OpenVLA	Long	−0.001	−0.003	0.3×
TracVLA	Goal	−0.058	−0.008	7.3×
TracVLA	Spatial	−0.031	−0.006	5.2×
TracVLA	Object	−0.047	−0.005	9.4×
TracVLA	Long	−0.036	−0.009	4.0×
NORA	Goal	−0.139	−0.007	19.9×
NORA	Spatial	−0.061	−0.005	12.2×

5.6 L1 Attention: A Base-Model Property

Table 12 reveals that L1 instruction-dominant head counts are *not* training-induced. TracVLA's untrained VLM (Phi-3-Vision) already has 32/32 instruction-dominant heads; training *reduces* this to 27–30/32. NORA's untrained Qwen2.5-VL has 13–14/16 heads; training preserves this. OpenVLA's OXE model has 24–25/32; LIBERO fine-tuning maintains similar counts (22–24/32), except Long OFT which reduces to 14/32.

This finding has an important implication: the number of instruction-routing heads at L1 reflects the backbone's language pretraining and visual-language alignment, not VLA task fine-tuning. The paper's core training-induced claims (R^2 collapse, CF sensitivity) are therefore orthogonal to L1 head counts.

OpenVLA Goal: L1 attention fractions. Table 13 shows that while head counts are similar, the trained model shows lower mean text attention (0.571 vs. 0.672) and comparable image attention (0.218 vs. 0.221), suggesting a redistribution toward mixed or action-focused processing.

6 Discussion

What does training create vs. inherit? Our results draw a clear line: R^2 exit collapse and CF sensitivity

Table 10: Per-architecture average exit slope.

Architecture	Tr. mean	Untr. mean	n
OpenVLA (Llama-2-7B)	−0.015	−0.003	4
TracVLA (Phi-3-mini)	−0.043	−0.007	4
NORA (Qwen2.5-VL-3B)	−0.100	−0.006	2

Table 11: Three-tier routing gradient (suggestive; Tier-2 $n = 1$).

Tier	Example	L1	CF	Coll.?
1: VLM	Phi-3/Qwen	28–32	cos .14–.17	No
2: OXE	openvla-7b	25/32	$d \approx 1.05$	No
3: Finetuned	All 3	13–30	$d \approx .8$–1.5	8/10

Table 12: L1 instruction-dominant heads: base-model-dependent.

Arch	Suite	Trained	Untrained	Δ
OpenVLA	Goal	24/32	25/32	−1
OpenVLA	Spatial	22/32	25/32	−3
OpenVLA	Object	24/32	24/32	0
OpenVLA	Long	14/32	25/32	−11
TracVLA	All 4	27–30	32/32	−2 to −5
NORA	Goal	13/16	13/16	0
NORA	Spatial	12/16	14/16	−2

Table 13: OpenVLA Goal: L1 attention fractions.

Metric	Trained	OXE	Δ
Instr-dom. heads	24/32	25/32	−1
Mean text attn	0.571	0.672	−0.101
Mean image attn	0.218	0.221	−0.003

are created by VLA fine-tuning (present in trained, absent in untrained regardless of architecture), while L1 head counts are inherited from the backbone's pretraining. This distinction matters for VLA design: optimizing entry-exit routing does not require starting from a specific number of instruction-dominant heads.

The Spatial CF inversion. OpenVLA Spatial's inverted CF ratio (0.81×) is the most challenging result. One interpretation: OXE's generalist sensitivity is partially overwritten by specialist fine-tuning. The model becomes more task-aligned but less broadly sensitive—trading breadth for depth. An alternative: Spatial tasks rely primarily on visual processing, so instruction routing contributes less, and fine-tuning primarily affects visual pathways. We cannot distinguish these interpretations with the current framework and flag this for future work.

Limitations.

1. Two of ten matrix cells show weak R^2 collapse, qualifying "architecture-independent" to "consistent across three tested architectures with caveats."
2. The three-tier gradient has only one Tier-2 datapoint (OXE/OpenVLA). Additional OXE-pretrained models would strengthen or refute this.
3. NORA evaluations cover only 2/4 suites. Object and Long checkpoints were unavailable.
4. CF ratios are not comparable across architectures due to baseline asymmetry. We mitigate this by reporting per-tier ratios, but a unified metric is needed.
5. All analysis is representational, not behavioral. We do not demonstrate that R^2 collapse or CF sensitivity predict downstream task performance. Behavioral validation is an important direction for future work.
6. OpenVLA Long uses OFT, confounding training method with architecture. A full fine-tuned Long checkpoint would clarify whether the weak collapse is method- or task-dependent.

7 Conclusion

We present a token-level analysis framework for VLAs and apply it across three architectures (OpenVLA, TracVLA, NORA) and four LIBERO task suites. R^2 exit collapse—the selective loss of action-relevant information in exit layers—emerges as a consistent training-induced signature (8/10 cells, $p = 0.002$), with steeper exit slopes in trained models (7.2× mean ratio vs. untrained). L1 instruction-dominant head counts, by contrast, are inherited from the backbone and not shaped by VLA fine-tuning. We observe a suggestive three-tier routing gradient from VLM base through general VLA to task-specific VLA, though the intermediate tier requires additional datapoints to confirm. Two exceptions (OpenVLA Spatial and Long/OFT) highlight that pretraining coverage and training method modulate the strength of exit collapse, motivating further study of the conditions under which this phenomenon does and does not emerge.

A Vision Encoder Geometry (Observation)

Under RSA, the vision encoder's internal geometry exhibits a non-monotonic pattern across VLA layers: an initial rise from input to mid-layers ($\Delta\rho \approx +0.016$, L0–L12), followed by decline ($\Delta\rho \approx -0.206$, L12–L32). Functional probes (R^2) confirm the mid-layer peak (L8–L13, $R^2 = 0.655$–0.678). Cross-encoder RSA rankings place DINOv2 highest ($\rho = 0.692$), above SigLIP+Projector (0.615) and CLIP ViT-L (0.557).

However, under CKA, the inverted-U pattern *does not replicate*: CKA shows monotonic increase, and DINOv2 ranks lowest. The RSA rise phase is small and may not

survive confidence interval analysis. We report this as a metric-dependent observation, not a core claim.

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